Cooperative control of a tethered drone and surface vessel for sustained presence at sea

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Declaration

I declare that this document is an original work of my own authorship and that it fulfills all the requirements of the Code of Conduct and Good Practices of the Universidade de Lisboa.

Contents

Acronyms

EMSA European Maritime Safety Agency
ILC iterative learning control
LQG linear quadratic Gaussian
LQR linear quadratic regulator
LTI linear time-invariant
MPC Model Predictive Control
NMPC Non-linear Model Predictive Control
PID proportional–integral–derivative
RL reinforcement learning
SISO single-input single-output
UAV Unmanned Aerial Vehicle
USV Unmanned Surface Vessel

Chapter 1

Introduction

1.1 Motivation

Saving lives at sea, protecting marine ecosystems, and securing maritime infrastructure all rely on one essential capability: understanding what is happening, where, and when. In vast and dynamic maritime environments, this requires maintaining reliable, high-resolution situational awareness over long periods. The European Maritime Safety Agency (EMSA) recognises this challenge and highlights the need for continuous monitoring and locally persistent operations to enable effective maritime surveillance and response [1].

Different maritime assets meet this challenge in complementary ways. Surface vessels give a physical presence at sea, with long endurance and large payload capacity [2, 3]. However, their range is limited by line of sight, and operations are expensive [4]. Aerial platforms, such as manned aircraft and Unmanned Aerial Vehicles (UAVs), allow fast deployment and elevated sensing [5]. Yet, aircraft are costly, and UAVs suffer from low endurance and unreliable communication [6].

Together, these assets form a connected system rather than separate solutions. Each one compensates for what the other lacks. Unmanned Surface Vessels (USVs) provides long-term presence at sea at much lower cost than crewed vessels, allowing wide and affordable patrols. UAVs extend this reach by sensing beyond the horizon from the air. Working as a team, they maintain continuous and local maritime awareness. This motivates the study of coordinated USV–UAV systems, where cooperation creates mission capabilities that neither platform can achieve alone in complex and changing environments.

A natural way to strengthen this cooperation is to add a physical tether between both platforms. The tether provides continuous power and reliable communication between the USV and UAV [7]. By placing the energy supply and communication systems on the USV, it overcomes two main limits of maritime UAVs: short endurance and weak long-range communication.

Figure 1.1 illustrates the configuration of the tethered USV–UAV system considered in this work.

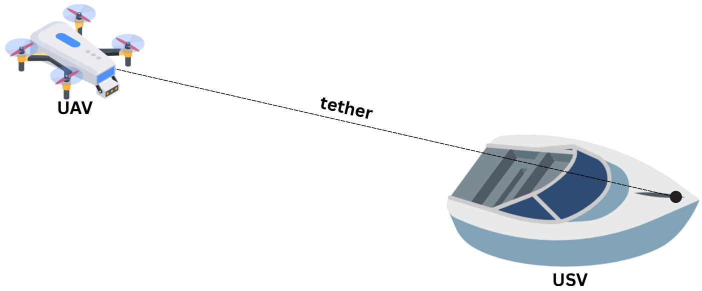


Figure 1.1: Schematic of a tethered USV–UAV system.

The main challenge of this work is to design a control framework that enables two autonomous vehicles, connected by a tether, to operate as a single coordinated system. The controller must manage not only the motion of both vehicles but also account for the tether constraints and dynamics, ensuring safe operation and robustness to wind, waves, and obstacles.

The USV and UAV operate under very different dynamics. The USV moves slowly, dominated by hydrodynamic drag and inertia, while the UAV reacts quickly, with motion evolving on much shorter time scales. Coordinated control is challenging because the controller must handle these different response rates and actuation dynamics consistently.

The maritime environment adds further complexity. Wind, waves, and water motion constantly disturb both platforms, making robustness to disturbances essential. Nearby structures, vessels, or obstacles can restrict their movement and risk tether entanglement. Effective cooperation relies on continuous information exchange and synchronised decision-making between both agents.

Most existing tethered UAV systems focus on power delivery and communication, with operation often remaining manual or semi-autonomous. In research prototypes, the aerial and surface vehicles are usually decoupled, the tether is simplified or linearised, or the surface platform is treated as fixed. These assumptions make control design easier but reduce realism. Few cooperative control approaches explicitly model the tether while coordinating both agents under realistic maritime conditions.

1.2 Problem Statement

This work addresses the problem of controlling a cooperative system composed of a USV and a tethered UAV operating in a coordinated manner. The objective of such a system is to autonomously execute maritime missions by following predefined waypoints or continuous reference trajectories, while respecting the physical and operational constraints of both platforms and of the tether that connects them.

To achieve this objective, the control framework must be capable of handling different phases of operation and mission requirements. This includes the definition of multiple control modes that allow the system to perform manoeuvres such as take-off and landing, as well as different forms of coordinated motion during navigation.

1.3 System Overview

Interaction with the system is defined at the mission level. The operator selects a mission, such as patrolling a given area, following a moving target, or visiting a set of waypoints that represent regions of interest. This mission information is received by a behaviour management layer, which selects the active control mode and handles mission logic, including switching between different relative configurations of the UAV with respect to the USV, as well as executing specific manoeuvres such as take off and landing.

Based on the selected mission and active control mode, the system generates coordinated motion for both the USV and the UAV. This process combines planning and control to compute trajectories and actuation commands that satisfy vehicle and tether constraints while pursuing the mission objectives. The UAV motion is designed to exploit its aerial perspective, providing sensing capabilities that complement those of the USV. The resulting control actions are applied directly to each vehicle, ensuring coordinated behaviour throughout the mission.

Figure 1.2 presents a high-level block diagram of the proposed system,

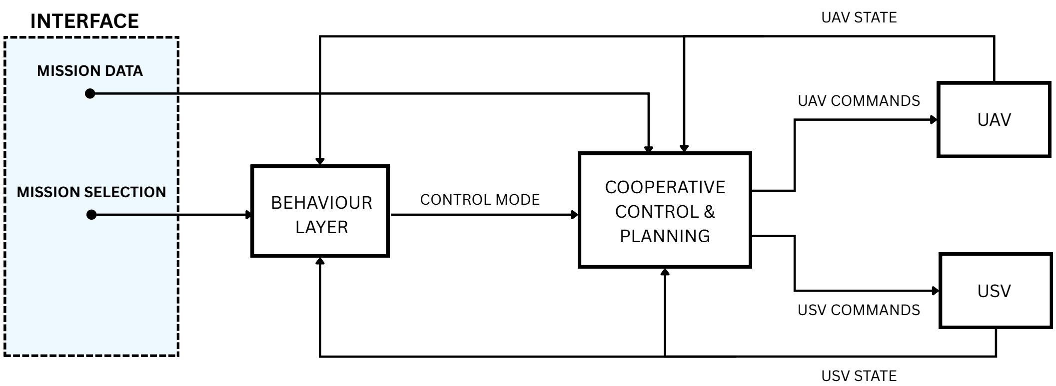


Figure 1.2: Proposed System Block Diagram

1.4 Expected Contributions

This thesis contributes to the design, implementation, and evaluation of cooperative control solutions for tethered UAV–USV systems, to enable autonomous and persistent maritime missions. The main contributions of this work are summarised as follows:

- A clear formulation of the cooperative control problem for a tethered UAV and an autonomous USV, including the main constraints and couplings introduced by the tether and by the maritime operating environment.

- The development of control-oriented models of the UAV, the USV, and the tether, capturing the dominant dynamics and interactions required for real-time control implementation.

- The design and implementation of a cooperative control framework capable of computing the individual actuation of both agents while ensuring coordinated behaviour across different mission phases.

- The definition and implementation of multiple control modes, such as area patrol, target following, take off, and landing.
- A simulation of the proposed cooperative control strategy under controlled and repeatable conditions.
- A systematic evaluation of system performance based on control-focused metrics, including tracking accuracy, control effort, robustness to external disturbances, and safe management of tether tension and geometry.

Chapter 2

State of the Art and Related Work

This chapter is structured into four main sections, each representing a key element of cooperative system control. Each section presents the main techniques commonly used in control theory, together with a subsection reviewing related work on topics closely related to this research.

The first section addresses *modelling and identification*, which provides the mathematical and experimental tools required to describe system dynamics and estimate relevant parameters. The second focuses on *control strategies*, ensuring stability, accurate tracking, and robustness to disturbances at the individual agent level. The third section concerns *cooperative coordination*, which enables communication and shared decision-making so that multiple agents can operate as a unified system. Finally, *path and trajectory planning* deals with the generation of feasible and safe motion plans that connect mission objectives with low-level control actions.

2.1 Modelling and Identification

Understanding how a system evolves is a prerequisite for any control design. Modelling provides this understanding by offering a mathematical description of the system dynamics, capturing the dominant behaviours while deliberately neglecting secondary effects that are not essential for analysis or control purposes [8]. The level of detail of such a model is inherently linked to the available physical insight, the amount and quality of experimental data, and the specific objectives for which the model is intended.

Within the control design process, an appropriate model forms the basis for analysing stability, performance, and robustness, both in linear and nonlinear settings. Rather than aiming for a replica of reality, modelling seeks a balance between fidelity and tractability, ensuring that the resulting description remains suitable for control synthesis and real-time implementation.

When first-principles knowledge is incomplete or difficult to formalise, system identification offers an alternative path by constructing models directly from measured input–output data. In this context, theoretical assumptions are complemented by experimental evidence, leading to a structured procedure that involves data collection, selection of an appropriate model structure, parameter estimation, and subsequent validation [9]. The quality of the identified model relies not only on the estimation algorithms, but also on careful experiment design and validation, often supported by frequency-domain analysis to assess consistency and predictive capability.

2.1.1 White-Box Modelling

White-box modelling is based on first principles and physical laws that govern system behaviour. By expressing the dynamics through analytical formulations, often derived from Newtonian mechanics, hydrodynamics, or aerodynamics, this approach provides clear insight into stability, controllability, and energy transfer.

To preserve analytical tractability, simplifying assumptions such as linearisation or the neglect of higher-order effects are commonly introduced. While these approximations facilitate analysis and control design, they can limit the ability of white-box models to represent nonlinear dynamics and uncertainties present in real-world systems.

2.1.2 Grey-Box Modelling

Grey-box modelling lies between purely analytical and purely data-driven approaches. The structure of the model is defined using physical insight, while parameters that are difficult to determine analytically, such as hydrodynamic or aerodynamic coefficients, are estimated from experimental data. This combination preserves a clear physical interpretation while improving accuracy through calibration.

The effectiveness of a grey-box model depends on how well the chosen structure represents the system and on the quality of the available data. For this reason, grey-box approaches are commonly used in control-oriented identification, where a balance between physical understanding and data-based tuning is essential for real-time estimation and adaptation.

2.1.3 Black-Box Modelling

Black-box modelling relies entirely on measured input–output data to describe system behaviour. Rather than using explicit physical laws, this approach identifies mathematical relationships that best fit the observed data. Typical examples include linear parametric models, state-space representations obtained from data, and learning-based methods.

This type of modelling is particularly useful when the underlying physics are complex, uncertain, or difficult to describe analytically. The added flexibility of black-box models comes at the cost of reduced interpretability and a stronger dependence on data quality. In cooperative robotic systems, such models are often used to represent coupled or nonlinear dynamics that are impractical to capture using first-principles models.

Once a model structure is defined, parameter estimation methods are used to infer the system dynamics from data.

2.1.4 Optimization-Based Estimation

Optimisation-based estimation focuses on identifying model parameters offline by minimising the mismatch between measured and predicted system outputs. This is typically achieved through cost functions based on least-squares formulations, including weighted least-squares and prediction-error methods (PEM) [9]. For nonlinear or constrained problems, advanced optimisation schemes such as trust-region methods and constrained least-squares are often employed, allowing grey-box models to be calibrated while enforcing physical bounds and regularisation. Frequency-domain approaches offer alternative formulations, particularly for systems affected by measurement noise or strong dynamic coupling.

2.1.5 Related Work

In systems connected by a tether, the model of the overall system must account at least for the tether geometry to constrain the relative motion of the agents and avoid physically infeasible configurations that could lead to tether failure.

In [10], a buoy–UAV system is presented in which the tether is modelled explicitly as a geometric and dynamic constraint between the buoy and the aerial vehicle. The tether tension is computed from the coupled dynamics and incorporated as an external force acting along the tether direction in the UAV model. This formulation relies on simplifying assumptions, namely that the tether is inextensible, massless, free of hydrodynamic effects, and remains taut during coupled operation.

An alternative modelling perspective is proposed in [11], where tether-induced forces are not included explicitly in the system dynamics. Instead, their effect is lumped into an external disturbance acting on the UAV. This disturbance is then handled at the control level through active disturbance rejection, avoiding the need for explicit tether modelling while still compensating for its influence in real time.

In [12], a cooperative UAV–USV system is modelled for the purpose of manipulating a floating object. Both vehicles are connected to the object through individual tethers, and the full system is described using simplified white-box models. The formulation assumes inextensible and massless tethers that remain permanently taut,

In [13], a tether is not included, and both the UAV and the USV are described using simplified white-box models. The USV is represented by a planar marine vehicle model commonly used for path-following applications, while the UAV is modelled independently using a reduced-order dynamic model.

2.2 Control Strategies

Control theory provides the mathematical framework for designing systems that regulate their behaviour through feedback. Its central goal is to ensure stability, accuracy, and robustness in the presence of uncertainty and disturbances. This section reviews the major families of control approaches, beginning with classical methods and progressing toward modern, predictive, and data-driven frameworks.

2.2.1 Classical Control

Classical control is based on the representation of systems through transfer functions in the Laplace or frequency domain, under the assumption of linear time-invariant (linear time-invariant (LTI)) dynamics.

Within this framework, feedback laws are typically implemented through proportional–integral–derivative (proportional–integral–derivative (PID)) controllers and frequency-domain compensators designed to shape the closed-loop response.

Despite its enduring importance, classical control was primarily developed for single-input single-output (single-input single-output (SISO)) systems and is typically based on approximate linear models. These limitations motivated the transition to state-space and multivariable formulations developed in the second half of the twentieth century. Nevertheless, classical techniques continue to provide the conceptual foundation of feedback design and remain an integral part of contemporary control education and practice.

2.2.2 Modern Control

Modern control theory generalises the principles of classical control to systems that are multivariable, time-varying, or subject to strong coupling between states and inputs. It is formulated in the *state-space* domain, where the internal dynamics of a system are described by sets of first-order differential equations rather than by input–output transfer functions. This representation enables systematic analysis of controllability, observability, and stability for both linear and nonlinear systems [14].

Within this framework, optimal control theory defines feedback laws that minimise a performance index, such as the linear quadratic regulator (LQR), its stochastic extension in the form of the linear quadratic Gaussian (LQG) controller, robust control methods such as H_∞ and μ-synthesis, and adaptive formulations [15–17].

2.2.3 Predictive and Optimisation-Based Control

Predictive and optimisation-based control methods determine control actions by solving an optimisation problem that predicts the system behaviour over a finite horizon. At each sampling instant, the controller minimises a cost function subject to system dynamics and operational constraints, typically penalising deviations from reference trajectories and excessive control effort. Only the first control action is applied, and the process repeats at the next time step, forming the receding-horizon principle that defines Model Predictive Control (MPC) [18].

MPC provides a systematic way to handle multivariable interactions and explicit constraints on inputs and states, which are often difficult to manage in traditional feedback designs. By embedding the system model within the optimisation, MPC achieves anticipatory behaviour, allowing control actions that account for predicted future disturbances and constraints [19, 20].

Extensions to nonlinear systems led to the development of Non-linear Model Predictive Control (NMPC), which replaces linear models with nonlinear formulations and uses numerical optimisation techniques to compute control inputs in real time. NMPC enables high-performance control for systems with strong nonlinearities or time-varying dynamics, while retaining the same predictive and constraint-handling advantages as its linear counterpart [19, 21]. The main challenge of predictive and optimisation-based control remains the computational effort required for online optimisation, which continues to drive research into efficient solvers and real-time implementations.

2.2.4 Data-Driven and Learning-Based Control

Data-driven and learning-based control methods aim to design controllers directly from data rather than from explicit physical or analytical models. Unlike model-based approaches, which rely on first-principles modelling or system identification, data-driven methods infer control laws or predictive models from measured input–output behaviour. This paradigm has gained traction due to advances in computation, optimisation, and machine learning, enabling adaptive control in complex or uncertain environments [22, 23].

Within this framework, two related approaches can be highlighted. iterative learning control (ILC) improves control performance in repetitive tasks by updating the control input after each execution [24]. reinforcement learning (RL) generalises this idea to uncertain or stochastic environments, optimising control policies through interaction with the system or its simulation [23, 25].

Recent advances combine machine learning techniques, such as neural networks, with control design to approximate complex nonlinear relationships between states, inputs, and actions. Purely data-driven controllers face challenges in stability, sample efficiency, and interpretability, motivating hybrid strategies that merge physics-based models with learned components. These model–learning combinations preserve analytical structure while improving adaptivity and represent one of the most active research directions in modern control [22, 26].

2.2.5 Related Work

In tethered UAV systems, the tether introduces additional forces and coupling effects that influence control design. In [10], this coupling is addressed through a nonlinear backstepping controller derived directly from the white-box dynamics of the UAV–buoy system. The work in [11] adopts an alternative approach based on active disturbance rejection, where the tether influence is not explicitly modelled but treated as an external disturbance estimated in real time by a reduced-order observer.

In [12], cooperative manipulation with two tethered agents is controlled using a MPC formulation based on simplified white-box models of the UAV, the USV, and the floating object. A similar predictive strategy is presented in [27], where both the UAV and the USV are coordinated using NMPC, relying on known dynamic models.

In [13], the USV employs a line-of-sight guidance law combined with PID control for heading and speed regulation. The UAV motion is controlled using an NMPC formulation, which generates reference accelerations to track the desired relative trajectory. No joint or centralised control law is employed, and each vehicle is controlled using its own dedicated controller.

2.3 Cooperative Coordination

Cooperative coordination refers to the control and organisation of multiple agents working towards achieving goals through communication and joint decision-making. In multi-vehicle systems, coordination ensures that agents act coherently while respecting their individual dynamics, sensing capabilities, and constraints. Fundamental cooperative behaviours include formation maintenance, consensus on shared states, cooperative manipulation, and coverage control [28, 29]. Depending on how information is exchanged and the control authority is structured, cooperative architectures are broadly categorised as *centralised* or *distributed*. The choice between them depends on mission objectives, communication reliability, and computational capacity [30, 31].

2.3.1 Centralized Coordination

In centralised coordination, a single decision node aggregates global information and computes control commands for all agents. This architecture facilitates global optimisation and ensures consistency of mission-level objectives, since the planner has access to full system states and constraints. Centralised schemes are particularly effective for coordinated trajectory planning, task allocation, and cooperative path optimisation, especially when communication links are stable, and delays are negligible [28, 31]. However, centralised control scales poorly with team size and is vulnerable to communication delays or single-point failures, which limit its applicability in large or uncertain environments.

2.3.2 Distributed Coordination

Distributed coordination eliminates the need for a central planner by allowing each agent to make decisions based on locally available information and limited communication with neighbours. Agents exchange state or intent data to achieve group-level objectives through consensus or distributed optimisation. This approach provides scalability and robustness to communication losses and agent failures, as decisions emerge from local interactions rather than a single control node [28, 32].

The theoretical foundation of distributed coordination is built on consensus theory, where agreement among agents is achieved via information exchange over network topologies. Graph-theoretic tools describe how connectivity and communication delays affect convergence and stability, forming the mathematical basis for distributed estimation, formation control, and cooperative task allocation [29, 33]. While distributed systems are more resilient and scalable, they face challenges related to synchronisation, delayed information, and ensuring stability in dynamically changing or heterogeneous networks [30].

2.3.3 Related Work

In [27], cooperation is based on a tracking strategy in which the UAV is controlled to follow and land on a moving USV. Only the UAV is actively controlled, while the USV moves independently and provides its state as a reference. The interaction is therefore unidirectional, with the UAV adapting its motion to the USV.

In contrast, [12] proposes a centralised model predictive control approach in which the UAV and the USV are controlled jointly. A single optimisation problem computes the control inputs for both vehicles, explicitly accounting for the coupling introduced by the tether. This approach focuses on cooperative object manipulation rather than on mission-level coordination between the vehicles.

In [13], coordination is achieved through a leader–follower structure. The USV acts as the leader and follows a predefined path, while the UAV adjusts its motion to maintain a desired relative configuration with respect to the USV. Coordination is therefore based on reference sharing and relative motion constraints, without joint optimisation or centralised decision-making.

2.4 Path and Trajectory Planning

Path and trajectory planning address the problem of generating feasible, collision-free, and dynamically consistent motions for autonomous systems operating in structured or unstructured environments. They constitute the intermediate layer between mission-level decision making and low-level control, translating abstract goals into trackable trajectories. The planning process must consider kinematic, dynamic, and environmental constraints to ensure safety, efficiency, and goal attainment [34, 35]. Methods are broadly divided into *offline* and *online* approaches, depending on when the plan is computed and how environmental information is handled. Offline planners typically exploit complete map data to compute globally optimal trajectories, while online planners focus on adaptability and real-time reactivity under uncertainty [36].

2.4.1 Offline Planning

Offline planning is performed before execution, assuming full or near-complete knowledge of the operating environment. Its objectives include global optimality, path smoothness, and completeness, that is, the guarantee of finding a feasible path if one exists. Classical approaches include graph-search methods such as Dijkstra's and A*, sampling-based methods such as Rapidly-exploring Random Trees and Probabilistic Roadmaps, and optimisation-based formulations such as trajectory optimisation and mixed-integer programming [34, 35]. Although these approaches can generate globally consistent motion plans, they are computationally demanding and rely on accurate models of the environment, which limits their practicality in dynamic or uncertain conditions.

2.4.2 Online Planning

Online planning addresses the generation and adaptation of trajectories in real time under dynamic or uncertain conditions. Unlike offline planners that rely on complete prior maps, online methods operate with partial and time-varying information, emphasising responsiveness and robustness over strict global optimality [35, 36]. Reactive techniques such as potential field methods, dynamic window approaches, and velocity obstacles provide fast obstacle avoidance by directly coupling perception and motion generation [37, 38]. However, these approaches may suffer from local minima or oscillatory behaviour in environments with multiple nearby obstacles.

To overcome these limitations, optimisation- and prediction-based methods integrate vehicle dynamics and constraints into the planning loop. MPC frameworks, in particular, enable online trajectory generation that accounts for system limits and environmental disturbances within a receding-horizon optimisation [18, 19]. The main challenges in online planning lie in maintaining computational tractability, ensuring stability under fast re-planning, and achieving reliable performance in multi-agent or highly nonlinear scenarios.

2.4.3 Related Work

A centralised predictive planning approach is presented in [12], where a single MPC formulation jointly optimises the motion of both the UAV and the USV. In this case, planning is achieved online through joint trajectory optimisation that explicitly accounts for the coupling introduced by tethers.

Planning is addressed from a different perspective in [13], where cooperative behaviour is achieved through coordinated path following. A predefined path is assigned to the USV, while the UAV generates its motion online as part of the control process to maintain coordination. In this framework, planning is limited to the online generation of the UAV motion, and no cooperative path planning between the two vehicles is performed.

Overall, existing works typically adopt one of two approaches to planning in UAV–USV systems, either a predefined path is assigned to one vehicle while the other generates its motion online to follow or support it, or both vehicles generate their motion online as part of a predictive control framework.

Chapter 3

Background

This chapter presents the theoretical and modelling foundations of the work, including an overview of the underlying concepts, the mathematical models of the system components, and the formulation of the control framework.

It is important to clarify the current scope of the work. Although the complete system comprises three coupled agents, a UAV, a USV, and a tether, the USV is not yet modelled or considered in the present stage. The focus is placed on a tethered UAV subsystem to enable the step-by-step development and validation of the proposed approach. The inclusion of the surface vessel within the same unified framework will be addressed in the full dissertation.

3.1 System Modelling

3.1.1 Modelling Principles

The objective of the system modelling in this work is to provide an accurate and control-oriented representation of the system dynamics, serving as the foundation for the motion control framework developed in subsequent sections. The adopted models aim to achieve a balance between physical accuracy and computational tractability, ensuring real-time feasibility while capturing the dominant dynamic effects governing the system behaviour [8].

Given the strong coupling introduced by the tether and the need for explicit dynamic consistency, the modelling approach initially follows a white-box methodology based on first-principles kinematic and dynamic formulations of the individual agents. This provides physically interpretable models with a clear structure for control design [39]. At later stages of the research, the framework may be extended towards a grey-box formulation by incorporating parameter estimation or simple data-driven refinements from experimental or simulated data to capture residual effects not fully represented in the physical model [9, 22, 40].

In general, the dynamic behaviour of a nonlinear control system can be expressed in continuous-time state-space form as [19, 39]

$$\dot{x} = f(x, u, t) + w, \tag{3.1}$$

where $x \in \mathbb{R}^n$ denotes the system state vector, $u \in \mathbb{R}^m$ the control input vector, and $w \in \mathbb{R}^n$ represents model uncertainties or external disturbances. This compact representation provides a unified mathematical structure for describing the system dynamics and forms the basis for the control formulation developed in later sections.

3.1.2 Dynamic Modelling of the tethered UAV Subsystem

Reference Frames and Notation

In this work, the UAV is modelled using a geometric formulation based on position vectors and rotation matrices [31, 39]. Linear and angular velocities (v, ω) are expressed in a body-fixed reference frame, while positions and trajectories are expressed in an inertial reference frame.

To describe the motion of the aerial vehicle relative to the Earth and its own orientation, two coordinate frames are defined, as illustrated in Fig. 3.1:

- An inertial reference frame $\{\mathcal{I}\}$, defined under a flat-Earth assumption.
- A body-fixed reference frame $\{\mathcal{B}\}$, rigidly attached to the UAV.

The configuration of the UAV is described by the pair (p, R), where:

- $p \in \mathbb{R}^3$ denotes the position of the origin of $\{\mathcal{B}\}$ expressed in $\{\mathcal{I}\}$.
- $R \in SO(3)$ represents the rotation matrix that transforms a vector expressed in the body-fixed frame $\{\mathcal{B}\}$ into its representation in the inertial frame $\{\mathcal{I}\}$.

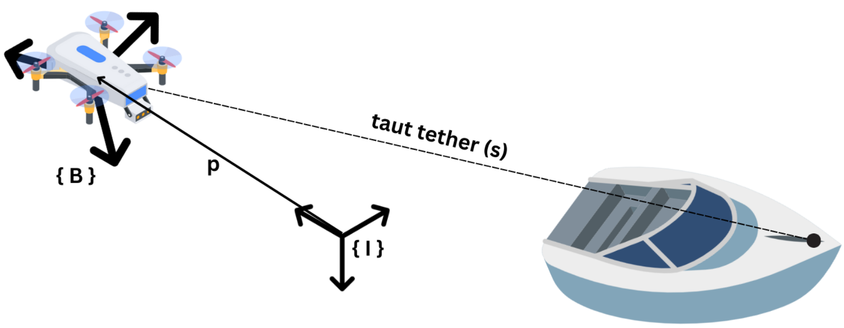


Figure 3.1: Inertial reference frame $\{\mathcal{I}\}$ and body-fixed reference frame $\{\mathcal{B}\}$ attached to the UAV, illustrating the position vector p and the relative orientation between frames.

Rigid Body Kinematics

The rigid-body kinematics describe how the position and orientation of the UAV evolve, relating the configuration variables (p, R) to the linear and angular velocities expressed in the body-fixed frame. The time evolution of the position, expressed in the inertial frame, is given by

$$\dot{p} = Rv, \tag{3.2}$$

where $v \in \mathbb{R}^3$ denotes the linear velocity expressed in the body-fixed frame.

The time evolution of the orientation is described by

$$\dot{R} = RS(\omega), \tag{3.3}$$

where $\omega \in \mathbb{R}^3$ denotes the angular velocity expressed in the body-fixed frame, and $S(\omega)$ is the associated skew-symmetric matrix defined as

$$S(\omega) = \begin{bmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{bmatrix}. \tag{3.4}$$

Rigid Body Dynamics

The rigid-body dynamics describe how the linear and angular velocities of the UAV evolve under the influence of external forces and moments. These equations are expressed in the body-fixed frame using the Newton–Euler equations. Let m denote the UAV mass and $J \in \mathbb{R}^{3\times3}$ its inertia matrix. The translational and rotational dynamics are given by

$$m\dot{v} = -S(\omega)\, mv + f, \tag{3.5}$$

$$J\dot{\omega} = -S(\omega)\, J\omega + n, \tag{3.6}$$

where $f \in \mathbb{R}^3$ denotes the resultant external force and $n \in \mathbb{R}^3$ the resultant external moment, both expressed in the body-fixed frame.

Quadrotor UAV Input Forces and Moments

The forces acting on the UAV include the thrust generated by the rotors, the gravitational force, and the force induced by the tether. While thrust and gravity are intrinsic to the vehicle, the tether introduces an additional coupling force that depends on the relative position between the UAV and its anchor point.

Thrust force and control moments The thrust is generated by the four rotors and acts along the body-fixed e_3 axis, where $e_3 = [0\ 0\ 1]^T$ denotes the unit vector aligned with the third axis of the body-fixed reference frame $\{\mathcal{B}\}$, as shown in Fig. 3.2. Each rotor produces an individual thrust T_i, and the total thrust magnitude is given by

$$T = \sum_{i=1}^{4} T_i. \tag{3.7}$$

The corresponding thrust force expressed in the body-fixed frame is

$$f_T = -Te_3, \tag{3.8}$$

In addition to the total thrust, differential rotor speeds generate control moments about the body axes. These moments arise from asymmetric thrust distributions and rotor reaction torques, and are collected in the control moment vector

$$n = [n_x \; n_y \; n_z]^T. \tag{3.9}$$

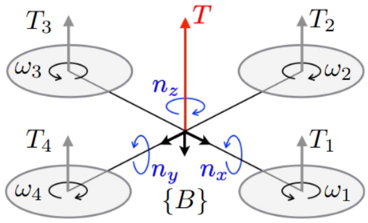


Figure 3.2: Body-fixed reference frame of the quadrotor showing the input forces and control moments generated by the rotors.

Gravitational force The gravitational force acts along the vertical direction of the inertial frame. When expressed in the body-fixed frame, it is given by

$$f_g = mgR^T e_3, \tag{3.10}$$

where m denotes the UAV mass and g the gravitational acceleration.

Tether-induced force The tether is modelled as a tension force acting along the line connecting the UAV to a fixed anchor point. Let $p \in \mathbb{R}^3$ denote the UAV position and $p_{\text{anchor}} \in \mathbb{R}^3$ the anchor point position, both expressed in the inertial frame. The relative position vector between the UAV and the tether anchor point is given by

$$s = p_{\text{anchor}} - p. \tag{3.11}$$

The tether is assumed to remain taut at all times, such that its direction and magnitude are determined solely by the relative position between the UAV and the anchor point. Under this assumption, the tether is modelled as a straight line connecting the two points, and slack conditions are not considered. The regularised distance is defined as

$$\|s\|_\varepsilon = \sqrt{s^T s + \varepsilon^2}. \tag{3.12}$$

The tether-induced force, expressed in the inertial frame, is defined as

$$F_{\text{tether}}(s) = (T_0 + w_c \|s\|_\varepsilon) \frac{s}{\|s\|_\varepsilon}, \tag{3.13}$$

where $T_0 > 0$ represents a minimum tension level that enforces continuous traction in the tether, $w_c > 0$ represents the effective weight of the cable per unit length, modelling the increase of required tension as the tether length grows, and $\varepsilon > 0$ is a small regularisation constant introduced to avoid numerical singularities when the relative distance becomes small.

This formulation is intended as a control-oriented approximation rather than a detailed structural model of a physical cable. The distance-dependent term reflects the intuitive effect that longer tethers require larger forces to remain taut due to the increasing weight of the cable.

When included in the UAV dynamics, the tether force is transformed to the body-fixed frame through the attitude rotation matrix and denoted by

$$f_{\text{tether}} = R^T F_{\text{tether}}. \tag{3.14}$$

Resultant external force The resultant external force acting on the UAV, expressed in the body-fixed frame $\{\mathcal{B}\}$, is given by

$$f = f_T + f_g + f_{\text{tether}}. \tag{3.15}$$

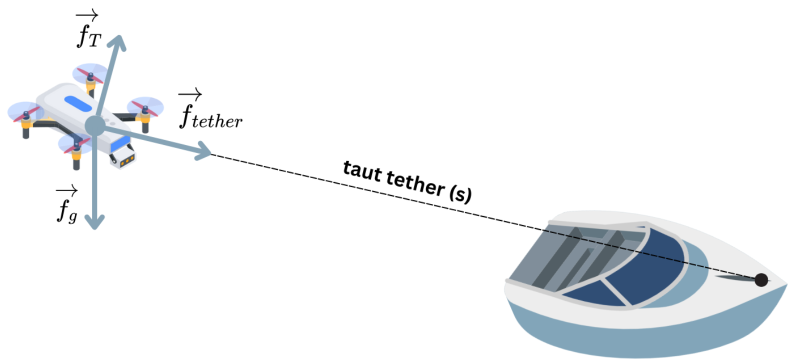


Figure 3.3: Forces acting on the tethered UAV: thrust $\vec{f}_T$ generated by the rotors, gravitational force $\vec{f}_g$, and tether tension $\vec{f}_{\text{tether}}$. All forces are expressed in the body-fixed frame $\{\mathcal{B}\}$.

Complete UAV Model

The complete dynamic model of the UAV is expressed in the body-fixed frame as

$$\begin{cases} \dot{p} = Rv, \\ \dot{R} = RS(\omega), \\ m\dot{v} = -S(\omega)\,mv - Te_3 + mgR^Te_3 + f_{\text{tether}}, \\ J\dot{\omega} = -S(\omega)\,J\omega + n, \end{cases} \tag{3.16}$$

where all forces are expressed in the body-fixed frame $\{\mathcal{B}\}$.

The translational dynamics can be equivalently expressed in the inertial reference frame by differentiating $\dot{p} = Rv$, yielding

$$m\ddot{p} = -TRe_3 + mge_3 + F_{\text{tether}}, \tag{3.17}$$

where $F_{\text{tether}} = Rf_{\text{tether}}$ denotes the tether force expressed in the inertial frame.

3.2 Motion Control

3.2.1 Control Architectures and Framework

The motion control problem of the UAV–USV system can be structured according to different architectures, depending on how coordination and control authority are distributed among agents [28, 29]. In a *distributed* architecture, each agent has its own controller, and cooperation arises through information exchange or consensus-based mechanisms. In a *centralised* formulation, the coupled system is treated as a single control entity, integrating all agent dynamics and constraints into one global optimisation problem.

Distributed control is computationally efficient and scalable, particularly when agent interactions are weak or communication delays are significant. These schemes rely on local control laws that use shared state information to achieve cooperative objectives such as formation keeping or consensus [29, 30]. However, in the tethered UAV–USV configuration considered here, the physical connection introduces strong dynamic coupling and geometric constraints across the system. Under such conditions, distributed control requires additional coordination or constraint enforcement layers, which can lead to suboptimal behaviour.

A centralised control formulation provides a unified representation of the entire system, enabling direct handling of all interactions and constraints within a single optimisation problem [12, 13]. This global view allows control actions that inherently satisfy tether-related and dynamic coupling effects.

Within this framework, nonlinear model predictive control (NMPC) is particularly suitable. NMPC computes control inputs by repeatedly solving a finite-horizon optimal control problem that predicts the system's evolution [19, 21]. This approach explicitly accounts for nonlinear dynamics, actuator limits, and coupling effects introduced by the tether. Although computationally demanding, NMPC yields globally consistent and dynamically feasible control actions. Consequently, a centralised NMPC strategy is adopted in this work as the foundation for the cooperative motion control framework.

3.2.2 Model Predictive Control Principles

MPC determines control actions not only from the current system state but also from predictions of its future evolution over a finite time horizon [18, 19]. By explicitly anticipating future behaviour, MPC can handle system dynamics and constraints proactively rather than reactively.

At each control instant t, the current state $x(t)$ is measured and used as the initial condition for a prediction of length N. Using a discrete-time model $x_{k+1|t} = f(x_{k|t}, u_{k|t})$, the future states $x_{k|t}$ are predicted for $k = t, \ldots, t + N$ based on a sequence of future control inputs $u_{k|t}$ [20]. An optimisation problem is then solved to determine the control sequence that best achieves the desired objectives, such as trajectory tracking or regulation, while satisfying physical and operational constraints. Only the first input $u_{t|t}$ is applied, and the process repeats at the next step, implementing the receding-horizon principle.

A key advantage of MPC is its ability to handle constraints explicitly. Bounds on states and inputs, such as actuator limits, safety margins, or geometric restrictions, are directly incorporated into the optimisation problem [21].

The general discrete-time formulation can be expressed as

$$\min_{u_{t|t},\dots,u_{t+N-1|t}} \left[\sum_{k=t}^{t+N-1} q(x_{k|t}, u_{k|t}) + p(x_{t+N|t}) \right], \tag{3.18}$$

subject to

$$x_{k+1|t} = f(x_{k|t}, u_{k|t}), \quad k = t, \dots, t+N-1, \tag{3.19}$$

$$x_{t|t} = x(t), \tag{3.20}$$

$$x_{k|t} \in \mathcal{X}, \quad k = t+1, \dots, t+N, \tag{3.21}$$

$$u_{k|t} \in \mathcal{U}, \quad k = t, \dots, t+N-1, \tag{3.22}$$

$$x_{t+N|t} \in \mathcal{X}_f, \tag{3.23}$$

where $x_{k|t}$ and $u_{k|t}$ denote the predicted state and input at step k given information at time t. The function $f(\cdot)$ represents the system dynamics, $q(\cdot)$ is the stage cost, and $p(\cdot)$ the terminal cost. The sets $\mathcal{X}, \mathcal{U}$, and $\mathcal{X}_f$ define admissible states, inputs, and terminal conditions, respectively.

When $f(\cdot)$ is nonlinear, the resulting NMPC formulation allows the controller to directly account for nonlinear dynamics and coupling effects, which are relevant in the tethered UAV–USV system considered in this work. Although the detailed stability and recursive feasibility conditions of NMPC are not discussed here, they will be further explored and analysed in the dissertation.

3.2.3 NMPC Formulation

Prediction Model

The control problem is formulated at the translational level by assuming a hierarchical control structure, which is standard in NMPC applications for aerial vehicles. The NMPC regulates the translational motion of the UAV, while the attitude dynamics are handled by a lower-level controller operating at a higher frequency.

Under this assumption, the translational dynamics expressed in the inertial reference frame, given in (3.17), are adopted as the basis for the prediction model. Dividing the inertial-frame dynamics by the vehicle mass and introducing a virtual control input corresponding to the acceleration generated by the thrust force, the translational dynamics used for prediction are written as

$$\ddot{p} = u_d + g e_3 + a_{\text{tether}}, \tag{3.24}$$

where

$$u_d = -\frac{T}{m} R e_3 \tag{3.25}$$

denotes the inertial acceleration generated by the thrust, and

$$a_{\text{tether}} = \frac{1}{m} F_{\text{tether}} \tag{3.26}$$

represents the acceleration induced by the tether.

State and Input Definition

Based on the prediction model above, the state vector considered by the NMPC is reduced to the translational state

$$x = \begin{bmatrix} p & \dot{p} \end{bmatrix}^T \in \mathbb{R}^6, \tag{3.27}$$

while the control input is given by the virtual acceleration

$$u = u_d \in \mathbb{R}^3. \tag{3.28}$$

NMPC Cost Function

The NMPC cost function is designed to track a desired translational trajectory while penalising excessive control effort. Over a prediction horizon of length N, the cost function is defined as

$$J = \sum_{k=1}^{N} \left[(p_k - p_k^{\text{ref}})^T Q_p (p_k - p_k^{\text{ref}}) + (\dot{p}_k - \dot{p}_k^{\text{ref}})^T Q_{\dot{p}} (\dot{p}_k - \dot{p}_k^{\text{ref}}) \right] + \sum_{k=0}^{N-1} u_k^T R u_k, \tag{3.29}$$

where Q_p, $Q_{\dot{p}}$, and R are positive definite weighting matrices that balance trajectory tracking accuracy and control effort.

The NMPC formulation also accounts for state and input constraints arising from physical, safety, and operational limitations of the tethered UAV. The explicit definition of these constraints is provided in Chapter 4.

Chapter 4

Implementation

This chapter describes the numerical implementation of the dynamic model and NMPC framework introduced in Chapter 3. It details the software environment, model discretisation, constraints, parameters and workflow used throughout this work.

4.1 Software Environment

The numerical implementation was developed in `Python`, which provides a flexible environment for rapid prototyping and integration of control and simulation components. The `CasADi` [41] framework is employed to define the system dynamics, cost function, and constraints symbolically, enabling automatic differentiation and the efficient formulation of the resulting nonlinear optimisation problem.

The nonlinear model predictive control problem is solved using the Interior Point OPTimizer (`IPOPT`) [42], a solver designed for large-scale nonlinear programming problems. Numerical operations and data handling are supported by the Python library `NumPy`, while `Matplotlib` is used for the visualisation of simulation results.

The overall implementation follows a modular structure, with separate components for model dynamics, cost definition, NMPC formulation, numerical simulation, and result visualisation. This organisation improves readability and facilitates future extensions of the framework.

4.2 Model Discretisation

The dynamic model introduced in Chapter 3 is formulated in continuous time. However, the NMPC algorithm computes control actions at discrete instants and relies on finite-horizon predictions. This requires a discrete-time representation of the system dynamics that can be evaluated repeatedly within the optimisation problem.

The continuous-time dynamics $\dot{x} = f(x, u_d)$ are discretised with a fixed sampling period $T_s = 0.05$ s, assuming a constant control input u_d within each interval.

To approximate the evolution of the state over each sampling interval, a fixed-step fourth-order Runge–Kutta (RK4) integration scheme is employed. The RK4 integration is implemented using CasADi's built-in numerical integrator, which automatically constructs the discrete-time state transition from the continuous-time dynamics.

Assuming the control input $u_{d,k}$ to be constant over the interval $[kT_s, (k+1)T_s]$, the RK4 scheme computes the state at the next time step as

$$x_{k+1} = x_k + \frac{T_s}{6}\left(k_1 + 2k_2 + 2k_3 + k_4\right),$$

where the intermediate evaluations are

$$k_1 = f(x_k, u_{d,k}), \quad k_2 = f\big(x_k + \tfrac{T_s}{2}k_1, u_{d,k}\big), \quad k_3 = f\big(x_k + \tfrac{T_s}{2}k_2, u_{d,k}\big), \quad k_4 = f(x_k + T_s k_3, u_{d,k}).$$

This procedure yields a fourth-order accurate numerical approximation of the continuous-time dynamics over one sampling interval. The discrete-time prediction model used within the NMPC can therefore be compactly expressed as

$$x_{k+1} = \Phi(x_k, u_{d,k}),$$

where $\Phi(\cdot)$ denotes the numerical state-transition map induced by the RK4 integration scheme.

4.3 Constraints

The NMPC formulation explicitly accounts for physical and operational constraints through inequality constraints included in the optimisation problem. These constraints are implemented directly at the prediction level and enforced at every time step along the prediction horizon.

In the implementation, all constraints are expressed in a smooth squared form, $\|\mathbf{x}\|^2 - x_{\max}^2 \leq 0$, to improve numerical stability and differentiability within the optimisation solver, while being mathematically equivalent to the standard norm formulation $\|\mathbf{x}\| \leq x_{\max}$.

4.3.1 Control Input Constraint

The magnitude of the virtual control input u_d, which represents the translational acceleration generated by thrust, is bounded as

$$\|u_{d,k}\| \leq u_{\max}.$$

This constraint limits the maximum thrust that can be generated by the UAV, $T \leq m\|u_d\|$.

4.3.2 Velocity Constraint

To prevent excessive translational velocities and to ensure dynamically feasible motion, a bound is imposed on the inertial velocity:

$$\|\dot{p}_k\| \leq v_{\max}.$$

4.3.3 Tether Length Constraint

The tether is assumed to remain taut at all times, implying that its effective length varies according to the relative position of the UAV with respect to the anchor point. In other words, the tether is allowed to retract or extend as the vehicle moves, while always maintaining tension.

Under this assumption, a maximum tether length constraint is imposed to represent the fully extended configuration of the cable. This geometric limitation is expressed as

$$\|p_k - p_{\text{anchor}}\| \leq L_{\max},$$

where p_k denotes the predicted position of the UAV at time step k, p_{anchor} is the fixed anchor point of the tether, and $L_{\max}$ represents the maximum cable length when the tether is fully deployed.

4.4 Parameters

The parameters adopted in the numerical implementation are selected exclusively for simulation purposes and are not associated with any specific hardware platform. Their primary objective is to enable an evaluation of the proposed NMPC framework.

In the full dissertation, these parameters will be revised based on the characteristics and constraints of a specific hardware setup and tuned based on performance metrics.

Table 4.1: Physical and control parameters used in the implementation.

Parameter	Symbol	Value
Mass	m	1.5 kg
Gravity	g	[0, 0, -9.81] m/s^2 (inertial frame, $z \uparrow$)
Tether base tension	T_0	3.0 N
Cable weight coefficient	w_c	0.2 N/m
Regularization constant	ε	0.02
Maximum tether length	$L_{\max}$	30 m
Sampling period	T_s	0.05 s
Prediction horizon	N	20 steps
Position weighting matrix	Q_p	$10I_3$
Velocity weighting matrix	$Q_{\dot{p}}$	$5I_3$
Control weighting matrix	R	$0.01I_3$
Maximum thrust acceleration	$u_{d,\max}$	$2g$
Maximum inertial velocity	$\dot{p}_{\max}$	5 m/s

4.5 NMPC Workflow

The NMPC problem is formulated symbolically using `CasADi`. The system dynamics, cost function, and constraints are assembled into a nonlinear program, which is solved at each control step using `IPOPT`. The following pseudocode summarises the overall closed-loop implementation workflow. It outlines the sequence of operations executed at each control step.

Algorithm 1 Closed-loop NMPC workflow

Define system model, cost function, and constraints
Formulate the NMPC problem over prediction horizon N
Initialise solver and set initial state x_0
while control loop is running **do**
    Update references and parameters
    Solve the NMPC optimisation problem
    Apply the first optimal control input $u_0^\star$
    Simulate system forward one sampling step to obtain x_{k+1}
    Update the NMPC problem with the new state and repeat
end while

Chapter 5

Results

This chapter presents the simulation scenarios considered in this work. The results obtained from the implementation presented in Chapter 4 are then shown and analysed, based on ideal simulations of the proposed system. As previously mentioned, the simulations and results analysed in this chapter are limited to the tethered UAV subsystem.

5.1 Simulation Scenarios

The objective is to validate the controller formulation and to assess its capability to track reference trajectories while explicitly enforcing the constraints introduced by the vehicle dynamics and by the tether.

Two simulation scenarios are considered. In the first scenario, a feasible reference trajectory is assigned to the UAV, respecting the imposed system constraints. This scenario is intended to evaluate the tracking performance of the controller and to analyse its ability to compensate for the dynamics introduced by the tether.

In the second scenario, the reference trajectory is deliberately defined such that it violates one or more system constraints. This case allows the evaluation of the controller's ability to correctly enforce these constraints, maintaining safe operation while tracking the reference trajectory whenever feasible.

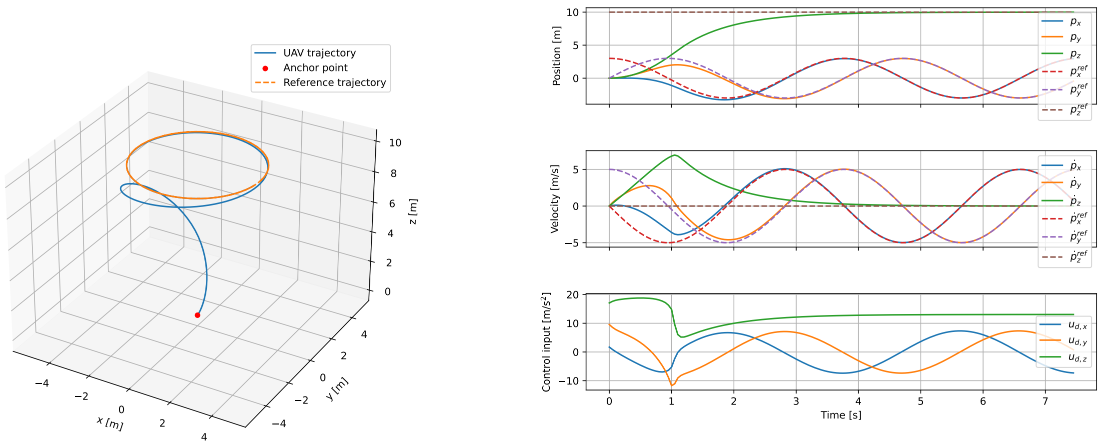
5.2 Nominal Scenario Results

In this scenario, a circular reference trajectory with a radius of 3 m and an altitude of 10 m was defined in the inertial reference frame. The inertial frame is oriented such that its z-axis points upward, facilitating the interpretation of the results.

The reference consists of a position vector $\mathbf{p}_{\mathrm{ref}}$ and a corresponding tangential velocity vector $\dot{\mathbf{p}}_{\mathrm{ref}}$ along the circular path, with a constant magnitude of 5 m/s. The UAV's initial state is located at the origin of the inertial frame, coinciding with the anchor point.

The reference trajectory is introduced into the nonlinear programming (NLP) problem as a time-varying parameter that evolves over the prediction horizon. Each horizon therefore contains a sequence of reference values, uniformly spaced by the sampling time T_s.

Figure 5.1(a) presents a 3D view of the UAV trajectory in the inertial frame, showing the transient phase from the initial position until the vehicle converges to and subsequently follows the prescribed circular path. Figure 5.1(b) illustrates the temporal evolution of the UAV states and the corresponding reference signals.



(a) UAV trajectory and reference path.

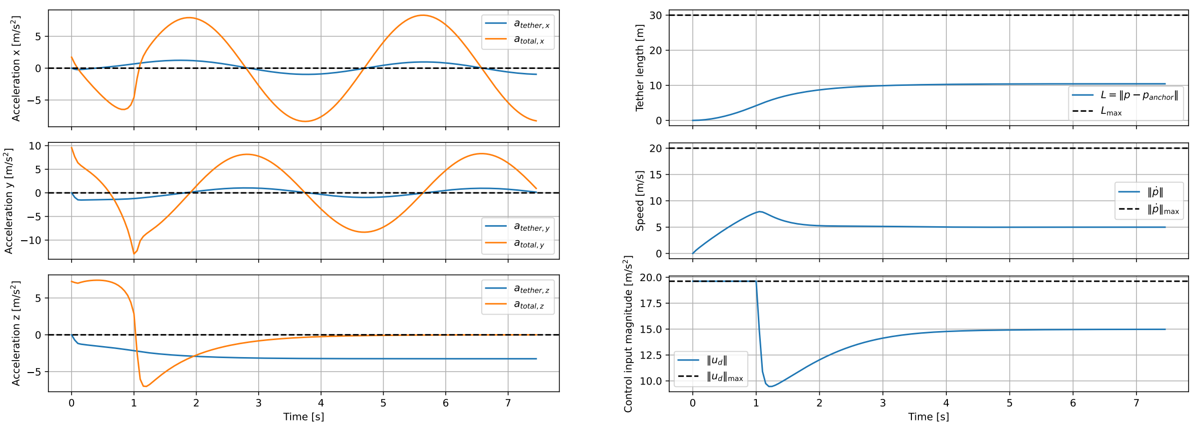
(b) Position, velocity, and control inputs with their corresponding references.

Figure 5.1: UAV trajectory, states, and control inputs for the nominal scenario.

The plots in Figure 5.1(a) evidence accurate tracking performance. Once the UAV reaches the desired altitude, all state variables converge to their respective references, as expected.

The inertial acceleration components $u_{d,(x,y,z)}$ responsible for this tracking performance show oscillations in the x and y components due to attitude variations that adjust the thrust direction and enable the UAV to follow the reference trajectory, while the z component stabilises at a value that compensates for gravity and the vertical force imposed by the tether.

Figure 5.2(a) shows the components of the acceleration induced by the tether on the UAV a_{tether}, as well as the total inertial acceleration $\ddot{\mathbf{p}}$, as a function of time. Figure 5.2(b) presents the three imposed constraints, the maximum tether length $L_{\max}$, the maximum velocity $\|\dot{\mathbf{p}}\|_{\max}$, and the maximum control acceleration $\|u_d\|_{\max}$, together with their respective constrained variables.



(a) Total and tether-induced acceleration components.

(b) Tether length, speed, and control acceleration magnitudes with respective constraints.

Figure 5.2: Acceleration and constraint magnitudes for the nominal scenario.

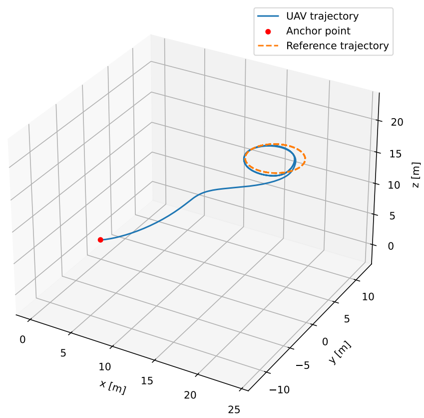
The z-component of the tether acceleration, $a_{\text{tether},z}$, increases as the tether extends from the anchor point until reaching the trajectory height of 10 m, since the tether is modelled as a force that grows with the UAV's distance from the anchor point. As expected, $\ddot{\mathbf{p}}_z$ converges to zero as the UAV maintains a constant altitude, while $\ddot{\mathbf{p}}_x$ and $\ddot{\mathbf{p}}_y$ exhibit oscillations due to the circular motion.

It can be observed that only the control acceleration magnitude, $\|u_d\| = T/m$, reached and respected its constraint of $2g$, stabilising once the UAV achieved a constant speed. This indicates that the thrust magnitude remained constant, while only its direction changed to follow the reference trajectory, as shown in Figure 5.1(b). The tether length L slightly exceeded 10 m, remaining well below its maximum value of 30 m, consistent with the defined trajectory, while the UAV's speed $\|\dot{\mathbf{p}}\|$ stabilised around 5 m/s, matching the reference.

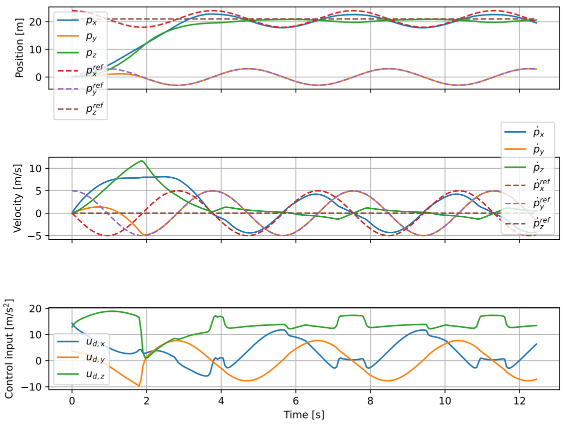
5.3 Constraint-Violating Scenario Results

In this scenario, the same circular trajectory was defined; however, its centre was shifted approximately 30 m away from the anchor point. As a result, part of the reference trajectory deliberately exceeded the reachable region imposed by the maximum tether length of 30 m, testing whether the constraint was properly enforced by the controller.

Figure 5.3(a) shows the UAV trajectory and the reference path for the constraint-violating scenario, while Figure 5.3(b) presents the corresponding state and control input evolution.



(a) UAV trajectory and reference path.

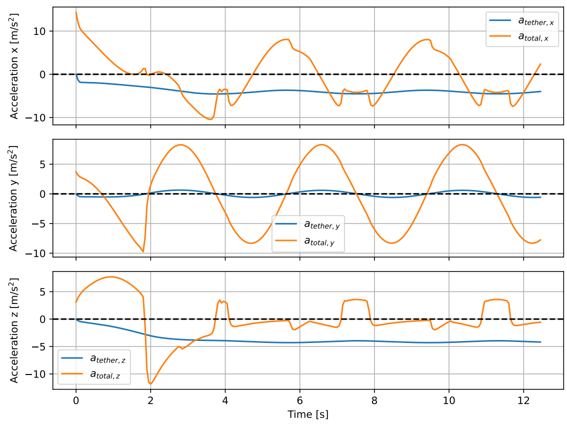


(b) Position, velocity, and control inputs with their corresponding references.

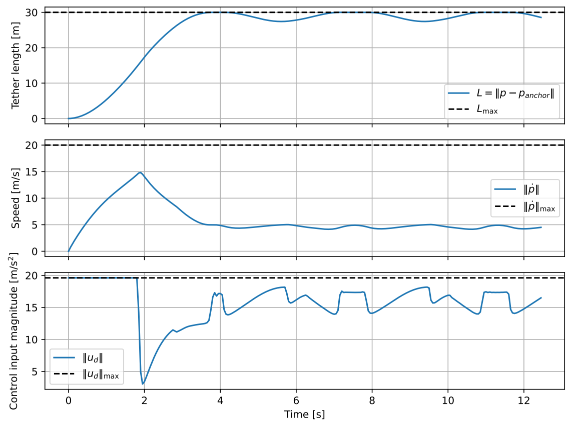
Figure 5.3: UAV trajectory, states, and control inputs for the constraint-violating scenario.

It can be observed that only a portion of the trajectory closer to the anchor point was accurately followed. Figure 5.3(b) shows that the UAV successfully sacrificed some tracking accuracy, particularly along the x and z axes, to satisfy the tether-length constraint. The corresponding control input $\mathbf{u}_d$ responsible for this motion reflects these adjustments, keeping a similar y component to the previous simulation and adapting the x and z components.

Figure 5.4(a) shows the total and tether-induced acceleration components for the constraint-violating scenario, while Figure 5.4(b) presents the corresponding constraint magnitudes.



(a) Total and tether-induced acceleration components.



(b) Tether length, speed, and control acceleration magnitudes with respective constraints.

Figure 5.4: Acceleration and constraint magnitudes for the constraint-violating scenario.

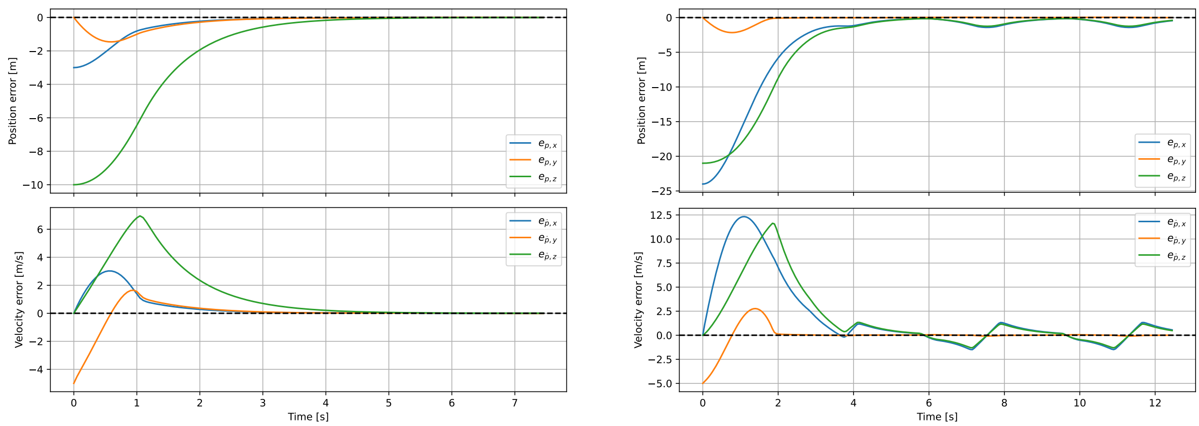
In Figure 5.4(a), the acceleration along the y-axis remains similar to that of the nominal scenario, while the x and z components adapt to the imposed constraint. Figure 5.4(b) clearly shows the tether length reaching its upper bound of 30 m without ever exceeding it, demonstrating that the constraint was correctly formulated and effectively enforced throughout the simulation.

5.4 Tracking Accuracy

The tracking performance of the controller is analysed by comparing the UAV states with their respective references, defined as

$$\mathbf{e}_p = \mathbf{p} - \mathbf{p}_{\text{ref}}, \qquad \mathbf{e}_{\dot{p}} = \dot{\mathbf{p}} - \dot{\mathbf{p}}_{\text{ref}}.$$

Figure 5.5(a) shows the tracking error for position and velocity components when the reference trajectory lies within the reachable zone, while Figure 5.5(b) presents the same quantities for the scenario where the reference trajectory extends beyond the tether's maximum length.



(a) Tracking error for the trajectory within the reachable zone.

(b) Tracking error for the trajectory outside the reachable zone.

Figure 5.5: Tracking error for position and velocity states.

As shown in Figure 5.5(a), after the UAV reaches the desired trajectory, approximately between $t = 5$ s and $t = 6$ s, the tracking errors converge to near-zero values, demonstrating the controller's effectiveness in accurately following the reference trajectory.

In Figure 5.5(b), as previously observed, the y component remains close to the reference, while the x and z components lose accuracy to maintain the trajectory within the tether's reachable region. This behaviour occurs because the trajectory centre was only shifted along the x and z axes, allowing the y component to achieve similar values to those of the nominal case while respecting the imposed constraints.

Chapter 6

Future Work and Timeline

This chapter outlines the work to be carried out during the dissertation period and presents a structured timeline for its development.

6.1 Future Work

This work established a basis for the dissertation through the implementation and ideal simulation of a tethered UAV controller. Building on this foundation, the framework will be extended to the full cooperative system and improved to allow testing under more realistic conditions.

A USV model will be developed, including an anchor point located on board. This model will be integrated into the NMPC formulation. The behaviour of the coupled system will be tested using ideal closed-loop simulations.

The model equations and control parameters will be refined for a specific UAV, USV, and tether hardware configuration. The system dynamics will also be extended to include time-varying external disturbances such as wind and waves. The resulting models will be validated and iteratively improved using simulation or experimental data. The refined and more complete models will then be integrated into the NMPC framework.

The convergence rate and optimality of the NMPC solutions will be analysed, with particular attention to feasibility and solver performance. A state observer based on an Extended Kalman Filter [43] may be used to combine model predictions with sensor measurements, providing more accurate and consistent state estimates for the NMPC. Different control modes will be tuned to handle distinct manoeuvres and mission phases.

The controller will be tested in non-ideal simulation environments using more accurate and realistic models. The performance will be evaluated in terms of tracking accuracy, stability, and disturbance rejection. The computational complexity of the NMPC optimisation problem will be analysed by assessing solver execution times and convergence behaviour, to determine whether the control problem can be reliably solved at the intended control update frequency, evaluating feasibility for real-time implementation.

6.2 Calendar

Figure 6.1 presents the main tasks planned for the dissertation and their corresponding timeline, organised in a Gantt chart [44].

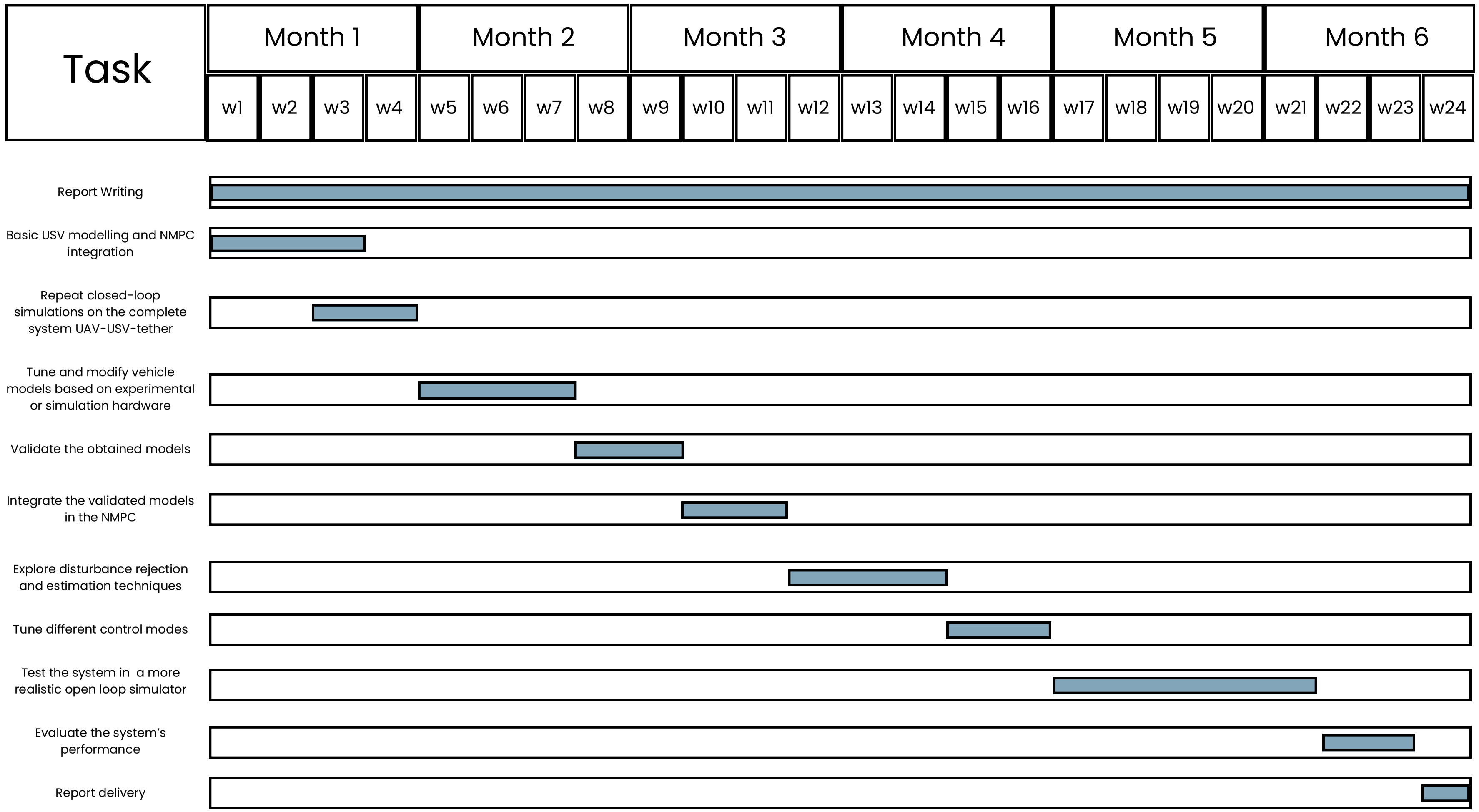


Figure 6.1: Gantt chart containing the task calendarisation for the dissertation period.

Bibliography

[1] E. M. S. Agency. EMSA Outlook 2025_v3.0, 2025.

[2] G. Briguglio and V. Crupi. Review on Sensors for Sustainable and Safe Maritime Mobility. *Journal of Marine Science and Engineering*, 12(2):353, Feb. 2024. ISSN 2077-1312. doi: 10.3390/jmse12020353.

[3] R. Dunley. Uncrewed naval vessels and the span of maritime tasks. *Marine Policy*, 149:105481, Mar. 2023. ISSN 0308-597X. doi: 10.1016/j.marpol.2023.105481.

[4] B. Şengül, F. Yılmaz, and Ö. Uğurlu. Safety–Security Analysis of Maritime Surveillance Systems in Critical Marine Areas. *Sustainability*, 15(23):16381, Jan. 2023. ISSN 2071-1050. doi: 10.3390/su152316381.

[5] N. Nomikos, P. K. Gkonis, P. S. Bithas, and P. Trakadas. A Survey on UAV-Aided Maritime Communications: Deployment Considerations, Applications, and Future Challenges. *IEEE Open Journal of the Communications Society*, 4:56–78, 2023. ISSN 2644-125X. doi: 10.1109/OJCOMS.2022.3225590.

[6] S. Bauk, N. Kapidani, L. Sousa, Ž. Lukšić, and A. Spuža. Advantages and disadvantages of some unmanned aerial vehicles deployed in maritime surveillance. *Maritime Transport Conference*, (8), 2020. ISSN 2938-8813.

[7] F. Fattori and S. Cocuzza. Tethered Drones: A Comprehensive Review of Technologies, Challenges, and Applications. *Drones*, 9(6):425, June 2025. ISSN 2504-446X. doi: 10.3390/drones9060425.

[8] C. W. de Silva. *Modeling of Dynamic Systems with Engineering Applications*. CRC Press, July 2022. ISBN 978-1-000-61439-8.

[9] A. K. Tangirala. *Principles of System Identification: Theory and Practice*. CRC Press, 2013. ISBN 978-1-040-08117-4.

[10] A. Kourani and N. Daher. Marine locomotion: A tethered UAV-Buoy system with surge velocity control. *Robotics and Autonomous Systems*, 145:103858, Nov. 2021. ISSN 09218890. doi: 10.1016/j.robot.2021.103858.

[11] G. Yu, J. Reis, D. Cabecinhas, R. Cunha, and C. Silvestre. Reduced-Complexity Active Disturbance Rejection Controller for Quadrotor-Slung-Load Transportation. *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, 53(8):5248–5259, 2023. ISSN 2168-2216. doi: 10.1109/TSMC.2023.3263881.

[12] F. Novák, T. Báča, and M. Saska. Collaborative Object Manipulation on the Water Surface by a UAV-USV Team Using Tethers. In *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 3425–3432, Oct. 2024. doi: 10.1109/IROS58592.2024.10802469.

[13] Z. Zhang, S. Liu, H. Wang, and H. Wu. A cooperative control framework for path following of UAV-USV cooperative system in maritime scenarios. *Ocean Engineering*, 342:122796, Dec. 2025. ISSN 00298018. doi: 10.1016/j.oceaneng.2025.122796.

[14] T. Kailath. *Linear Systems*. Prentice-Hall Information and System Sciences Series. Prentice-Hall, Englewood Cliffs, NJ, 1980. ISBN 978-0-13-536961-6.

[15] B. D. O. Anderson and J. B. Moore. *Optimal Control: Linear Quadratic Methods*. Prentice Hall Information and System Sciences Series. Prentice Hall, Englewood Cliffs, NJ, 1. [print.] edition, 1990. ISBN 978-0-13-638560-8.

[16] K. Zhou, J. C. Doyle, K. Glover, and J. C. Doyle. *Robust and Optimal Control*. Prentice Hall, Upper Saddle River, NJ, 1996. ISBN 978-0-13-456567-5.

[17] K. J. Åström and B. Wittenmark. *Adaptive Control: Karl Johan Åström and Björn Wittenmark, Lund Institute of Technology*. Dover Publications, Mineola, second edition edition, 1995. ISBN 978-0-486-31914-8 978-1-306-32427-4.

[18] E. F. Camacho and C. Bordons. *Model Predictive Control*. Advanced Textbooks in Control and Signal Processing. Springer, London, second edition edition, 2007. ISBN 978-1-85233-694-3 978-0-85729-398-5. doi: 10.1007/978-0-85729-398-5.

[19] J. B. Rawlings, D. Q. Mayne, and M. Diehl. *Model Predictive Control: Theory, Computation and Design*. Nob Hill Publishing, LLC, Santa Barbara, 2nd edition edition, 2020. ISBN 978-0-9759377-5-4.

[20] D. Mayne, J. Rawlings, C. Rao, and P. Scokaert. Constrained model predictive control: Stability and optimality. *Automatica*, 36(6):789–814, June 2000. ISSN 00051098. doi: 10.1016/S0005-1098(99)00214-9.

[21] L. Grüne and J. Pannek. *Nonlinear Model Predictive Control: Theory and Algorithms*. Communications and Control Engineering. Springer-Verlag London Limited, London, 2011. ISBN 978-0-85729-500-2 978-0-85729-501-9. doi: 10.1007/978-0-85729-501-9.

[22] Z.-S. Hou and Z. Wang. From model-based control to data-driven control: Survey, classification and perspective. *Information Sciences*, 235:3–35, June 2013. ISSN 00200255. doi: 10.1016/j.ins.2012.07.014.

[23] B. Recht. A Tour of Reinforcement Learning: The View from Continuous Control. *Annual Review of Control, Robotics, and Autonomous Systems*, 2(1):253–279, May 2019. ISSN 2573-5144, 2573-5144. doi: 10.1146/annurev-control-053018-023825.

[24] D. A. Bristow, M. Tharayil, and A. G. Alleyne. A survey of iterative learning control. *IEEE Control Systems*, 26(3):96–114, June 2006. ISSN 1066-033X, 1941-000X. doi: 10.1109/MCS.2006.1636313.

[25] R. S. Sutton and A. Barto. *Reinforcement Learning: An Introduction*. Adaptive Computation and Machine Learning. The MIT Press, Cambridge, Massachusetts London, England, second edition edition, 2018. ISBN 978-0-262-03924-6.

[26] M. Korda and I. Mezić. Linear predictors for nonlinear dynamical systems: Koopman operator meets model predictive control. *Automatica*, 93:149–160, July 2018. ISSN 00051098. doi: 10.1016/j.automatica.2018.03.046.

[27] W. Li, Y. Ge, Z. Guan, H. Gao, and H. Feng. NMPC-based UAV-USV cooperative tracking and landing. *Journal of the Franklin Institute*, 360(11):7481–7500, July 2023. ISSN 00160032. doi: 10.1016/j.jfranklin.2023.06.023.

[28] W. Ren and R. Beard. *Distributed Consensus in Multi-Vehicle Cooperative Control: Theory and Applications*. Springer, 2010. ISBN 978-1-84996-701-3.

[29] R. Olfati-Saber, J. A. Fax, and R. M. Murray. Consensus and Cooperation in Networked Multi-Agent Systems. *Proceedings of the IEEE*, 95(1):215–233, Jan. 2007. ISSN 0018-9219. doi: 10.1109/JPROC.2006.887293.

[30] Y. Cao, W. Yu, W. Ren, and G. Chen. An Overview of Recent Progress in the Study of Distributed Multi-Agent Coordination. *IEEE Transactions on Industrial Informatics*, 9(1):427–438, Feb. 2013. ISSN 1551-3203, 1941-0050. doi: 10.1109/TII.2012.2219061.

[31] R. W. Beard. *Small Unmanned Aircraft: Theory and Practice*. Princeton University Press, Princeton, 2012. ISBN 978-0-691-14921-9 978-1-4008-4060-1.

[32] R. Olfati-Saber and R. Murray. Consensus Problems in Networks of Agents With Switching Topology and Time-Delays. *IEEE Transactions on Automatic Control*, 49(9):1520–1533, Sept. 2004. ISSN 0018-9286. doi: 10.1109/TAC.2004.834113.

[33] M. Mesbahi and M. Egerstedt. *Graph Theoretic Methods in Multiagent Networks*. Princeton Series in Applied Mathematics. Princeton University Press, Princeton, N.J, course book edition, 2010. ISBN 978-1-4008-3535-5. doi: 10.1515/9781400835355.

[34] H. M. Choset. *Principles of Robot Motion: Theory, Algorithms, and Implementations*. Intelligent Robotics and Autonomous Agents Ser. MIT Press, Cambridge, 2005. ISBN 978-0-262-03327-5 978-0-262-25591-2.

[35] S. M. LaValle. *Planning Algorithms*. Cambridge University Press, 1 edition, May 2006. ISBN 978-0-521-86205-9 978-0-511-54687-7. doi: 10.1017/CBO9780511546877.

[36] B. Siciliano. *Springer Handbook of Robotics*. Springer Handbooks Ser. Springer International Publishing AG, Cham, 2nd ed edition, 2016. ISBN 978-3-319-32552-1.

[37] D. Fox, W. Burgard, and S. Thrun. The dynamic window approach to collision avoidance. *IEEE Robotics & Automation Magazine*, 4(1):23–33, Mar. 1997. ISSN 10709932. doi: 10.1109/100.580977.

[38] P. Fiorini and Z. Shiller. Motion Planning in Dynamic Environments Using Velocity Obstacles. *The International Journal of Robotics Research*, 17(7):760–772, July 1998. ISSN 0278-3649, 1741-3176. doi: 10.1177/027836499801700706.

[39] J.-J. E. Slotine and W. Li. *Applied Nonlinear Control*. Prentice Hall, Englewood Cliffs, NJ, 1991. ISBN 978-0-13-040890-7 978-0-13-040049-9.

[40] L. Ljung. *System Identification: Theory for the User*. Pearson Education, Dec. 1998. ISBN 978-0-13-244053-0.

[41] J. A. E. Andersson, J. Gillis, G. Horn, J. B. Rawlings, and M. Diehl. CasADi: A software framework for nonlinear optimization and optimal control. *Mathematical Programming Computation*, 11(1):1–36, Mar. 2019. ISSN 1867-2949, 1867-2957. doi: 10.1007/s12532-018-0139-4.

[42] A. Wächter and L. T. Biegler. On the implementation of an interior-point filter line-search algorithm for large-scale nonlinear programming. *Mathematical Programming*, 106(1):25–57, Mar. 2006. ISSN 0025-5610, 1436-4646. doi: 10.1007/s10107-004-0559-y.

[43] D. Simon. *Optimal State Estimation: Kalman, H∞, and Nonlinear Approaches*. Wiley, 1 edition, May 2006. ISBN 978-0-471-70858-2 978-0-470-04534-3. doi: 10.1002/0470045345.

[44] H. L. Gantt. *Organizing for Work*. Hive Publishing Company, 1974. ISBN 978-0-87960-046-4.